

ORF307: Optimization

Precept: #4

Agenda

- Approximate solutions of linear equations with multiple RHS
- Varying the RHS in linearly constrained Least Squares
- Nearest vector with given average
- Multi-objective control (see section 15.2 in Boyd & Vandenberghe book [VMLS])
- Rendezvous (see Python notebook in Companion)

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Approximate solution of linear equations with multiple RHS's

- Suppose A is invertible
- We seek x such that $Ax \approx b_i$ for $i = 1, 2, \dots, k$

Example: In many real-world scenarios, we need to find a single solution that satisfies multiple, slightly different, linear equations. Let's see if two seemingly different approaches to this problem yield the same result.

- Bob proposes a *multi-objective* least squares approach

$$x^{Bob} = \arg \min_x \|Ax - b_1\|_2^2 + \dots + \|Ax - b_k\|_2^2$$

- Alice proposes to first average the b_i 's to get $b = (b_1 + \dots + b_k)/k$ and then choose x by solving the system of linear equations

$$Ax^{Alice} = b$$

- Oscar says that the two approaches sound different, but actually we have:

$$x^{Bob} = x^{Alice}$$

- Is Oscar right?

Solution:

Let's compute Bob's and Alice's solutions, and then try to see if Oscar is right or not.

- Alice's approach

$$\begin{aligned} Ax^{Alice} = b &\implies Ax^{Alice} = \frac{1}{k}(b_1 + \dots + b_k) \implies A^T Ax^{Alice} = \frac{1}{k}A^T(b_1 + \dots + b_k) \implies \\ x^{Alice} &= \frac{1}{k}(A^T A)^{-1}A^T(b_1 + \dots + b_k) \end{aligned}$$

- Bob's approach

- Stack the matrices A 's and the b_i 's into a block matrix $\tilde{A}$ and a larger vector $\tilde{b}$

$$\tilde{A} = \begin{bmatrix} A \\ \vdots \\ A \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_k \end{bmatrix},$$

- Bob's problem can be written in the compact form (looks like a *single-objective* least squares problem)

$$x^{Bob} = \arg \min_x \|\tilde{A}x - \tilde{b}\|_2^2$$

- Normal equations:

$$\begin{aligned} \tilde{A}^T \tilde{A}x^{Bob} &= \tilde{A}^T \tilde{b} \implies [A^T \dots A^T] \begin{bmatrix} A \\ \vdots \\ A \end{bmatrix} x^{Bob} = [A^T \dots A^T] \begin{bmatrix} b \\ \vdots \\ b \end{bmatrix} \implies \\ \implies \left(\sum_{i=1}^k A^T A \right) x^{Bob} &= \sum_{i=1}^k A^T b_i \implies k(A^T A)x^{Bob} = A^T \sum_{i=1}^k b_i \implies \\ \implies x^{Bob} &= \frac{1}{k}(A^T A)^{-1}A^T \sum_{i=1}^k b_i \end{aligned}$$

- Oscar's claim is true! As we have shown, both approaches lead to the exact same solution.

Varying the RHS in *constrained* linear least squares

- Consider the constrained linear least squares problem

$$\begin{aligned}\hat{x} &= \arg \min_x \|Ax - b\|_2^2 \\ &s.t. \quad Cx = d\end{aligned}$$

- Assume the KKT matrix is invertible
- Think of $\hat{x}$ as a function of b and d
- Our friend Oscar asserts that $\hat{x}$ is a *linear function* of b and d and hence must have the form

$$\hat{x} = Fb + Gd$$

for some appropriate matrices F and G .

- Is Oscar right? If yes, justify and explain how to find F and G . If not, give a counter-example.

Solution:

We will form the KKT optimality conditions we saw in the lectures:

$$\begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix} \begin{pmatrix} \hat{x} \\ \hat{z} \end{pmatrix} = \begin{pmatrix} 2A^T b \\ d \end{pmatrix} \implies \begin{pmatrix} \hat{x} \\ \hat{z} \end{pmatrix} = \begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix}^{-1} \begin{pmatrix} 2A^T b \\ d \end{pmatrix}$$

Define the matrix

$$H = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} = \begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix}^{-1}$$

Then we have

$$\begin{pmatrix} \hat{x} \\ \hat{z} \end{pmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{pmatrix} 2A^T b \\ d \end{pmatrix} = \begin{pmatrix} 2H_{11}A^T b + H_{12}d \\ 2H_{21}A^T b + H_{22}d \end{pmatrix}$$

Hence $\hat{x} = 2H_{11}A^T b + H_{12}d$ is indeed a linear function of b and d with $F = 2H_{11}A^T$ and $G = H_{12}$.

Oscar is right, again.

Remark: This linearity means that if we know the optimal solutions for two different pairs of (b,d) , we can find the solution for any linear combination of those pairs **without having to re-solve** the entire optimization problem.

Nearest vector with a given average

- Consider the vector $a \in \mathbb{R}^n$ and the scalar $\beta \in \mathbb{R}$
- Find $x \in \mathbb{R}^n$ such that it is closest to vector a among all n -vectors that have average value β .

Solution:

We will formulate this problem as constrained linear least squares problem:

$$\begin{aligned} \min \quad & \|x - a\|_2^2 \\ \text{s.t.} \quad & \mathbf{1}^T x = n\beta \end{aligned}$$

Recall the general definition of the KKT conditions for the constrained linear least squares

$$\begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix} \begin{pmatrix} \hat{x} \\ \hat{z} \end{pmatrix} = \begin{pmatrix} 2A^T b \\ d \end{pmatrix}$$

If we substitute $A = I, b = a, C = \mathbf{1}^T$ and $d = n\beta$ we obtain the KKT condition of our problem

$$\begin{bmatrix} 2I & \mathbf{1} \\ \mathbf{1}^T & 0 \end{bmatrix} \begin{pmatrix} \hat{x} \\ \hat{z} \end{pmatrix} = \begin{pmatrix} 2a \\ n\beta \end{pmatrix} \implies \begin{aligned} 2\hat{x} + \mathbf{1}\hat{z} &= 2a \\ \mathbf{1}^T \hat{x} &= n\beta \end{aligned}$$

Then we solve the first block for $\hat{x}$

$$\hat{x} = a - \frac{1}{2}\mathbf{1}\hat{z} \tag{1}$$

and substitute it to the second block

$$\begin{aligned} \mathbf{1}^T(a - \frac{1}{2}\mathbf{1}\hat{z}) &= n\beta \implies \mathbf{1}^T a - \frac{1}{2}\mathbf{1}^T \mathbf{1}\hat{z} = n\beta \implies \\ \implies \mathbf{1}^T a - \frac{1}{2}n\hat{z} &= n\beta \implies \hat{z} = \frac{2}{n}(\mathbf{1}^T a - n\beta) \implies \hat{z} = 2(\text{avg}(a) - \beta) \end{aligned}$$

Substituting $\hat{z}$ into (1) we get

$$\hat{x} = a - \frac{1}{2}\mathbf{1}2(\text{avg}(a) - \beta) \implies \hat{x} = (a - \mathbf{1}\text{avg}(a)) + \mathbf{1}\beta$$

In words, to find the closest vector $\hat{x}$ to the known vector a , we simply need to subtract the average of the elements of the vector a from every element of a , and then add β to every element.

Example: Consider

$$a = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix} \quad \text{and} \quad \beta = 4.$$

Then $\text{avg}(a) = 6$, and we can easily compute

$$\hat{x} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix} - \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix} + \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix}$$

We see that, component-wise, the distance between $\hat{x}$ and a is the same (and equal to 2).

Remark: As the result shows, the solution simply shifts the vector a so that its average matches β , while preserving the relative distances between its components.